Aeronautics and Space Technology Club

IIT (ISM) Dhanbad

Signal Processing Ladder: A Hands-On Problem Series from Sensors to Machine Learning

Problem Statement 17 · Part D: Digital Signal Processing Level 3

Track: STFT, Spectrogram, Wavelets, Instantaneous Frequency, Chirps in Noise

Prerequisites: Problems 12, 13 · **Estimated effort:** 14–18 hours

Time–Frequency Analysis: When the Spectrum Changes with Time

A satellite's Doppler curve, an engine run-up and a gravitational-wave inspiral are all signals whose frequency moves. Build the STFT from scratch, choose the window by the chirp-rate rule, whiten noisy data, track a chirp ridge, fit the inspiral law to recover a chirp mass, compare with wavelet scalograms, and apply the method to open gravitational-wave data.

General Instructions

- **Thorough Review:** Read the whole problem statement, including the theory, before writing any code. Do the short analytic derivations *by hand first*; the code is there to confirm and extend your understanding, not to replace it.
- **Pacing:** This is a self-paced problem. The split below is a guide for the effort estimate in the header.

Stage	Phase	What you should have at the end of it
1	Think (∼20%)	Theory understood; every analytic result derived; signals, units and sampling rates written down.
2	Build (∼30%)	Core code working end to end on the baseline data and validated against the analytic results.
3	Experiment (∼30%)	Full sweeps, stress tests and the real or synthetic datasets processed; first plots and numbers.
4	Refine (∼20%)	Weaknesses fixed, honest analysis and memo written, repository folder clean.

- **Compute and Tools:** Everything is designed to run on a laptop CPU in Python 3.10+ (NumPy, SciPy, Matplotlib, and where stated pandas, Astropy, scikit-learn, PyWavelets and PyTorch). No GPU and no hardware are required; where a problem offers an optional hardware experiment it is marked as a bonus. A full run should never need more than a few minutes; if yours does, vectorise before you scale up.
- **Data:** Each problem lists (i) a *synthetic generator* from `sp_common.synth` (offline, seeded, with known ground truth) and (ii) *real datasets* to download yourself. Develop and validate on the synthetic data first, then repeat the key result on real data. Never commit large raw data files to the repository.
- **Submission Format:** Push your work to `solutions/17_time_frequency_analysis/` in the repository: (i) clean, commented, reproducible Jupyter notebook(s) with fixed random seeds, (ii) a short written report (`report.pdf` or `report.md`) containing the requested analysis sections, and (iii) all figures in a `figures/` sub-folder. The report must be understandable without running the code.

- **Originality:** All code, analysis and written explanations must be your own work. Any external library, snippet, paper or AI tool you use must be cited in the report. You should be able to explain every line you submit.
- **Honesty over Hype:** Validate against analytic results *before* trusting a number. A modest or negative result that is correctly analysed is worth more than an inflated claim. Report error bars, false-alarm rates and failure cases, not only your best run.
- **Units:** Write the units of every quantity and state in the first notebook cell whether a frequency is in Hz or rad/s and whether it is normalised to the sampling rate. Most mistakes in signal processing are factors of 2π, factors of 2 in one-sided spectra, or an off-by-one in the frequency axis.
- **Getting Stuck:** Re-read the theory section, check the self-check targets, then ask your mentor or open an issue in the repository. Describe what you expected, what you got, and what you already tried.

Background and Motivation

The Fourier transform answers "which frequencies are present" but not "when". For stationary signals that is enough; for the signals that matter most in aerospace it is not. The received tone of a passing satellite sweeps through 10 kHz in minutes; a rocket engine's vibration lines sweep as the shaft spins up and cross structural resonances; the gravitational wave from two merging black holes sweeps from 35 Hz to 250 Hz in 0.2 s.

The short-time Fourier transform (STFT) slides a window through the data and computes a spectrum in each position. It is limited by the uncertainty principle of problem 04: a short window localises events in time but smears frequency, a long window does the reverse. Wavelets use windows that scale with frequency, which suits transients. In noisy data a spectrogram is only useful after *whitening*: dividing the data by the amplitude spectral density of the noise so that every frequency contributes equally.

Theoretical Framework

1. STFT and Spectrogram

$$X[m,k] = \sum_n x[n]\, w[n-mH]\, \mathrm{e}^{-\mathrm{j}2\pi kn/N}, \qquad \mathrm{SPEC}[m,k] = |X[m,k]|^2, \tag{1}$$

with hop H and window w. The inverse STFT by overlap-add is exact if the windows satisfy the constant-overlap-add (COLA) condition $\sum_m w[n-mH] = \text{const}$; for a periodic Hann window it holds for hops of $N/2$ and $N/4$. Time resolution $\approx$ window length, frequency resolution $\approx \mathrm{ENBW} \cdot f_s/N_w$.

2. Chirps and Instantaneous Frequency

A linear chirp $x(t) = \cos\left(2\pi(f_0 t + \tfrac{1}{2}kt^2)\right)$ has instantaneous frequency $f(t) = f_0 + kt$ (rate k in Hz/s). The spectrogram ridge follows $f(t)$ best when the window length is about $T_w \approx 1/\sqrt{k}$ (the frequency change within a window equals the window's resolution). For a monocomponent signal the analytic signal $x_a = x + \mathrm{j}\mathcal{H}\{x\}$ gives $f(t) = \frac{1}{2\pi}\mathrm{d}\arg x_a/\mathrm{d}t$.

3. Inspiral Chirp

For two compact objects with chirp mass $\mathcal{M}_c$ at leading (Newtonian) order, with $T_c = G\mathcal{M}_c/c^3$ and time to coalescence $\tau = t_c - t$,

$$f(\tau) = \frac{1}{\pi}\left(\frac{5}{256\,\tau}\right)^{3/8} T_c^{-5/8}, \qquad \Phi(\tau) = -2\left(\frac{\tau}{5T_c}\right)^{5/8}, \qquad h \propto f^{2/3}\cos\Phi, \tag{2}$$

so $f^{-8/3} = \pi^{8/3}\frac{256}{5}T_c^{5/3}\,(t_c - t)$ is a *straight line* in time; its slope gives $\mathcal{M}_c$, its zero-crossing the merger time.

4. Whitening

If the noise has one-sided PSD $S_n(f)$, the whitened data $\tilde{d}(f) = d(f)/\sqrt{S_n(f)}$ has a flat spectrum. In practice S_n is estimated by Welch's method on a stretch without signal (problem 13), and the data is band-passed to the sensitive band.

5. Wavelets

The continuous wavelet transform with a mother wavelet ψ is $W(a,b) = a^{-1/2}\int x(t)\psi^*\big((t-b)/a\big)\mathrm{d}t$. For the Morlet wavelet $\psi(t) = \pi^{-1/4}\mathrm{e}^{\mathrm{j}\omega_0 t}\mathrm{e}^{-t^2/2}$ the scale a corresponds to the frequency $f \approx \omega_0/(2\pi a)$ (in units of the sampling interval). Windows are long at low frequency and short at high frequency (constant Q). The discrete wavelet transform (DWT) with dyadic filter banks splits the band into octaves: level j detail coefficients cover roughly $[f_s/2^{j+1},\ f_s/2^j]$.

Data and Simulation Specification

Linear chirp	$f_s = 4\,\mathrm{kHz}$, $f_0 = 200\,\mathrm{Hz}$, $k = 100\,\mathrm{Hz/s}$, duration 4 s, white noise at amplitude SNR 1 and 0.3.
Gravitational-wave-like chirp	`sp_common.synth.chirp_in_noise(fs=4096, duration=16, t_merger=12, mchirp_msun=30, target_snr=12)`: chirp from 20 Hz to about 148 Hz in 0.75 s in coloured noise; also with `target_snr=8` and 25.
Doppler pass	`sp_common.synth.satellite_doppler_pass(duration_s=120, fs=20000, snr_db=20)` and with `snr_db=0`.
Run-up (create yourself)	Shaft speed ramping from 600 to 6000 rpm in 20 s with harmonics 1–4 and a structural resonance at 250 Hz.
Real data (strongly recommended)	GW150914: 32 s of LIGO Hanford strain at 4 kHz around GPS time 1126259462 from the Gravitational Wave Open Science Center (GWOSC, use the `gwosc`/`gwpy` tutorials). Alternative: a SatNOGS waterfall of a CubeSat pass.

Your Task

1. **Window length and the uncertainty trade-off (Stage 2).** Compute spectrograms of the clean linear chirp for window lengths $T_w = 0.02,\ 0.1,\ 0.5\,\mathrm{s}$. Measure the ridge width in time and frequency, show the optimum near $1/\sqrt{k} = 0.1\,\mathrm{s}$ and explain with Eq. (1) and the uncertainty principle.
2. **STFT from scratch.** Implement the STFT and inverse STFT (overlap-add) with a Hann window. Verify perfect reconstruction for 50% and 75% overlap, show failure for a hop that violates COLA, compare with `scipy.signal.stft` and express the spectrogram in correct PSD units (problem 12).

3. **Chirp in coloured noise.** Take `chirp_in_noise`. Show the raw spectrogram (the chirp is invisible), estimate the PSD by Welch's method from a stretch without signal, whiten and band-pass to 20 Hz–300 Hz, and show the whitened spectrogram. Extract the ridge and report the frequency-versus-time track with a measure of its scatter at SNR $= 8, 12, 25$.
4. **Chirp-law fit.** Plot $f^{-8/3}$ against time for your ridge, fit a straight line and use its slope to estimate $\mathcal{M}_c$ and its zero-crossing to estimate the merger time. Compare both with the truth and report the uncertainty (bootstrap over ridge points); state how it depends on SNR.
5. **Continuous wavelet transform.** Implement the Morlet CWT with FFT-based convolution. Compute scalograms for the same data and for a short transient (a 10-ms sine-Gaussian blip at 200 Hz plus the chirp), compare with the spectrogram on a log-frequency axis and describe where each representation is superior.
6. **Hilbert instantaneous frequency and Doppler.** Estimate the instantaneous frequency of the clean Doppler pass with the analytic signal; at 0 dB SNR use the spectrogram ridge. Fit the straight-line model $\Delta f = -f_0 v^2 t/(c\sqrt{d^2 + v^2 t^2})$ for d, v and t_0.
7. **Discrete wavelet analysis and a Campbell diagram.** Decompose the chirp record with a Daubechies-4 DWT (`pywt`), plot the per-level energy and identify the levels containing the chirp. Then build the run-up signal, compute its spectrogram, overlay the engine-order lines and identify the time at which each harmonic crosses the 250 Hz resonance.
8. **Real data.** Process GW150914 (or your SatNOGS pass): whiten with a Welch PSD from off-source data, band-pass 35 Hz–350 Hz, plot the spectrogram or Q-transform and read the frequency track. Compare the observed sweep with the published behaviour (a rise from about 35 Hz to 250 Hz in roughly 0.2 s).

Self-check targets

For $k = 100\,\text{Hz/s}$ the optimal window is $1/\sqrt{k} = 0.1\,\text{s}$. A periodic Hann window with hop $N/2$ or $N/4$ reconstructs perfectly to machine precision. The synthetic chirp runs from 20 Hz to about 148 Hz; $f^{-8/3}$ is linear in time with negative slope and crosses zero at the merger time $t_c = 12\,\text{s}$. DWT level j covers $[f_s/2^{j+1}, f_s/2^j]$.

Aerospace Application Link

During a static test of a rocket engine turbopump the shaft speed ramps from rest to operating speed while a structural mode sits at a fixed frequency. Using your Campbell diagram, state at which speeds each engine order crosses the mode, estimate the amplification of vibration at each crossing from the spectrogram, and explain how a faster run-up (shorter dwell at resonance) trades against thrust-transient loads. Describe how the same tools on a CubeSat's Doppler curve give the time of closest approach and the pass geometry for orbit determination.

Deliverables

D1 **Window trade-off:** spectrogram comparisons and measured resolutions.

D2 **STFT implementation:** the code, the perfect-reconstruction tests and the COLA failure.

D3 **Chirp in noise:** whitening pipeline, whitened spectrograms and the ridge extraction at three SNRs.

D4 **Chirp-mass fit:** the linearised fit with uncertainties versus SNR, and the CWT/DWT comparison.

D5 **Doppler and Campbell diagram:** fit parameters with errors, and the resonance-crossing table.

D6 **Real-data report and memo (300 words):** GW150914 (or SatNOGS) time–frequency analysis and "what does the frequency track tell us?" Cite at least two external sources.

Bonus (a) Compute the Wigner–Ville distribution and explain cross-terms. (b) Implement the constant-Q (Q-)transform used in LIGO glitch studies. (c) Use the synchrosqueezing transform to sharpen the ridge.

Learning Resources

Time–Frequency Methods

- Cohen, L. *Time–Frequency Analysis.*
- Mallat, S. *A Wavelet Tour of Signal Processing.*
- Torrence, C., Compo, G.P. (1998). "A practical guide to wavelet analysis." Bull. Am. Meteorol. Soc. 79, 61.
- Allen, J.B., Rabiner, L.R. (1977). "A unified approach to short-time Fourier analysis and synthesis." Proc. IEEE 65, 1558.

Gravitational-Wave Data and Chirps

- Abbott, B.P. et al. (LIGO and Virgo) (2016). "Observation of gravitational waves from a binary black hole merger." Phys. Rev. Lett. 116, 061102, arXiv:1602.03837.
- Gravitational Wave Open Science Center, data and tutorials: `https://gwosc.org`; GWpy: `https://gwpy.github.io`
- Maggiore, M. *Gravitational Waves, Vol. 1*, Ch. 4 (the inspiral chirp).

Tools

- SciPy `signal.stft`, `signal.spectrogram`, `signal.hilbert`; PyWavelets: `https://pywavelets.readthedocs.io`; SatNOGS: `https://db.satnogs.org`

Evaluation

Score your own submission out of 100 on five equally weighted criteria, then ask your mentor to score it independently and compare.

Criterion	What to look for	Marks
Conceptual depth	Correct mathematics and physics; every result explained through a mechanism (why the spectrum looks like this), not only through code behaviour.	/20
Implementation quality	Correct, modular, commented, reproducible code; no unexplained magic numbers; analytic checks and self-check targets pass.	/20
Aerospace-application integration	The signal-processing result is carried through to a measurable consequence for a sensor, a link, a flight system or a detection decision, as the task requires.	/20
Analysis and critical thinking	Error bars, convergence, comparison to theory, honest treatment of limitations, false alarms and failure cases.	/20
Communication	Clear report, labelled plots (units, legends, captions) and a memo that a non-expert student could follow.	/20

Bonus tasks are not part of the 100 marks. A bonus attempted badly does not cost you marks, and a well-executed one is the best sign that you own the material.